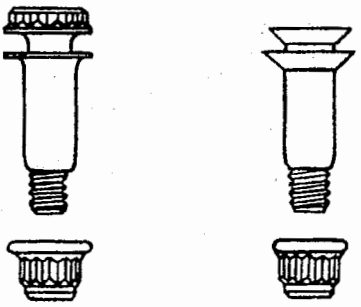
MIL-HDBK-271-104
15 NOVEMBER 1985

SLEEVEBOLT
INSTALLATION HANDBOOK

MILITARY HANDBOOK

AIRCRAFT FASTENERS, INSTALLATION

AND REMOVAL METHODS FOR



MIL-B-8831 and MIL-B-85667
Expandable Straight Shank Bolts
(Sleevebolts)

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DEPARTMENT OF DEFENSE
WASHINGTON 25 D.C. 20361

MIL-HDBK-271-104

AIRCRAFT FASTENERS, INSTALLATION AND REMOVAL METHODS FOR MIL-B-85667 AND MIL-B-8831 SLEEVEBOLTS

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.

2. The publication was approved on 15 November 1985 for printing and inclusion in the military standardization handbook series.

3. This handbook provides installation and removal methods for sleevebolts. It will provide valuable information and guidance to aircraft mechanics at Navy, Army and Air Force Rework Facilities.

   This handbook is not intended to be referenced in procurement specifications except for information purposes, nor shall it supersede any specification requirements.

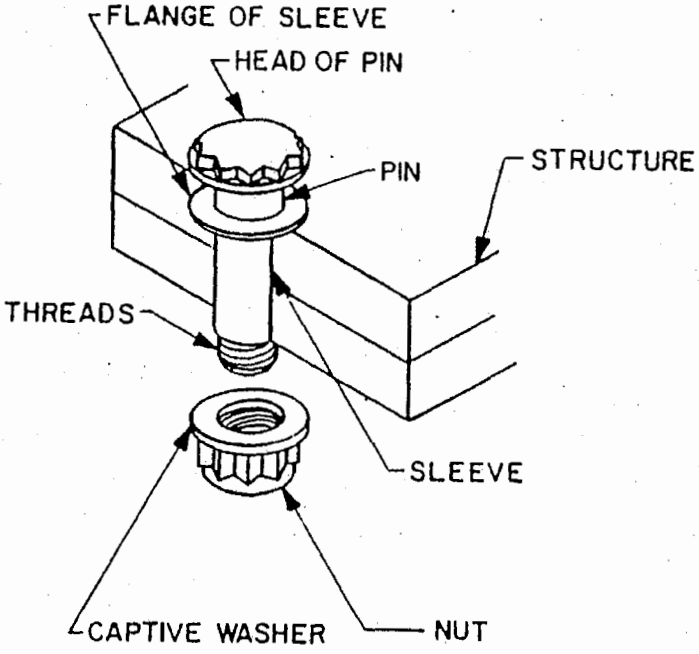
4. Every effort has been made to include the latest information on installation and removal methods of sleevebolts. Mechanics should review this handbook periodically. Users of this handbook are encouraged to report any errors or recommendations to: Commanding Officer, Naval Air Engineering Center, Code 93, SESD, Lakehurst, NJ 08733.

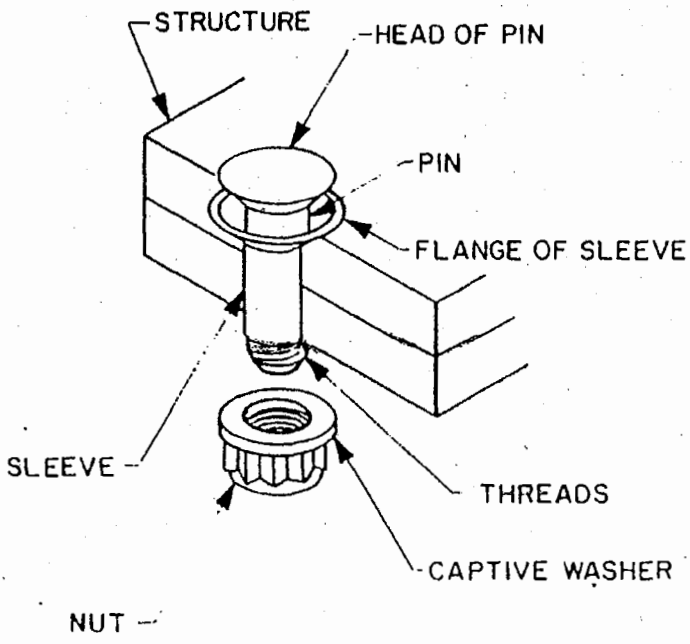
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EXPANDABLE STRAIGHT SHANK BOLT ("SLEEVEBOLT SYSTEM")

SCOPE:

Sleevebolts may be used in design, repair and modification. These bolts may be used in lieu of tapered or straight shank interference fit fatigue rated fasteners.





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EXPANDABLE STRAIGHT SHANK BOLT SLEEVEBOLT FASTENER

INTRODUCTION:

The purpose of this handbook is to cover the proper installation and removal of sleeve-bolts. This fastening system is an assembly consisting of a headed pin and a flanged sleeve, ground externally straight for use in straight-sided clearance holes. These fasteners are specified in MIL-B-85667 and MIL-B-8831. Interference fit in proportion to the bolt protrusion from the sleeve is realized as the bolt head underside surface comes in contact with the sleeve flange which is in contact with the structure. These bolts are used in lieu of tapered-shank fasteners (Taper-loks) when tooling or field conditions become a problem with tapered shank bolts.

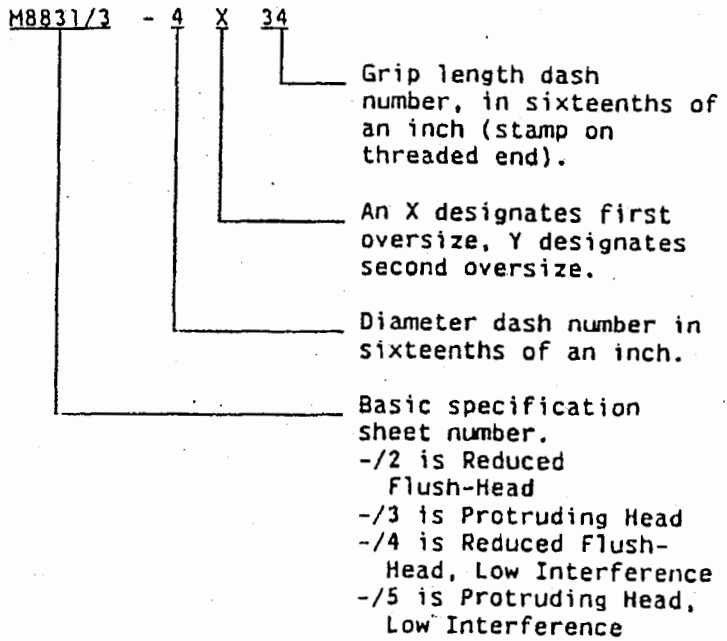
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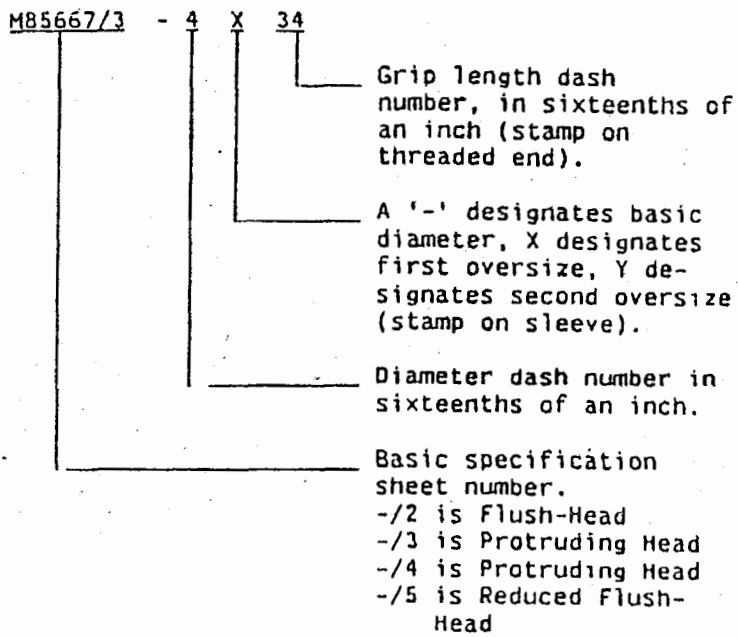
FASTENER PART NUMBER:

Part number coding for Expandable-Shank Fasteners (Sleevebolt):

EXAMPLE OF A PART NUMBER FOR FASTENERS SPECIFIED BY MIL-B-8831:



EXAMPLE OF A PART NUMBER FOR FASTENERS SPECIFIED BY MIL-B-85667:



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COMPANION NUTS FOR SLEEVEBOLTS

Nuts for MIL-B-85667/2 and /4 bolts shall be in accordance with MIL-N-85729.

Nuts for MIL-B-85667/3 and /5 bolts shall be in accordance with MIL-N-85730.

Nuts for all MIL-B-8831 bolts shall be in acordance with MIL-N-8985/3 or MIL-N-8922/1.

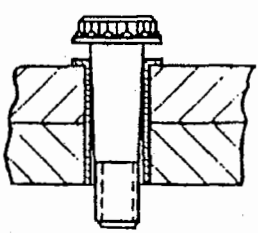
EXPANDABLE-SHANK FASTENER GAGES:

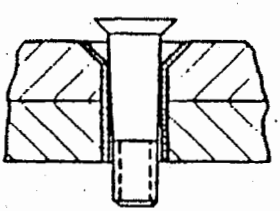
The actual bolt is used as a gage following a simple procedure.

A.
Check for proper fillet-radius, making sure sleeve flange is flush with structure

(protruding head & flush-head).

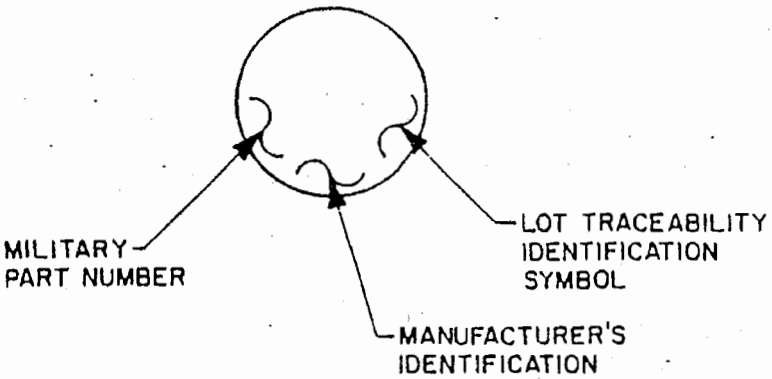
B.
By drilling and checking depth with the sleevebolt, obtain the proper countersink depth (flush-head type).





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HEAD MARKING:



GRIP SELECTION:

On flush-head type bolts, the grip is measured from the top of the sleeve flange to the end of the sleeve. On protruding head type bolts, the grip is measured from the underside of the sleeve flange to the end of the sleeve.

These bolts are designed for use in stack-ups equal to but not less than .062 inches. The extra length of the bolt in a short stack-up may be accommodated using a washer nut with a recessed counterbore.

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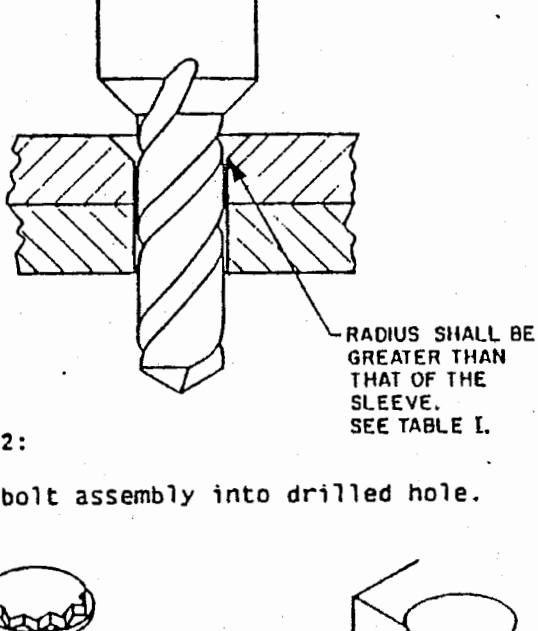
INSTALLATION:

Step # 1:

Drill straight diameter hole.

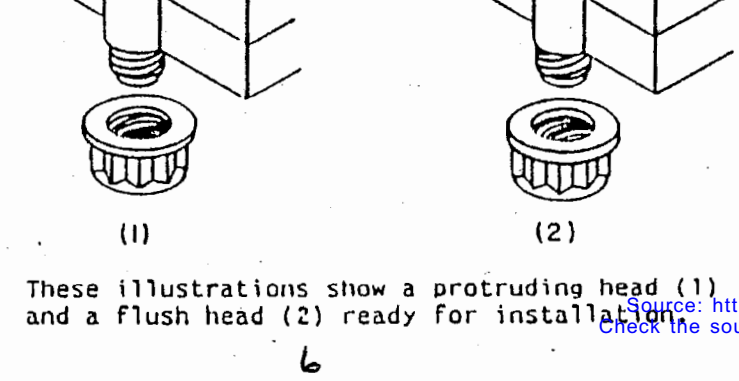
A. Select drill sizes for flush head bolts from Table I.

B. Select drill sizes for protruding head bolts from Table II.



Step # 2:

Insert bolt assembly into drilled hole.



(1) (2)

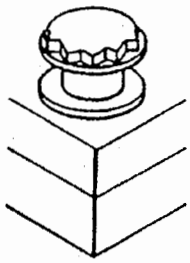
These illustrations show a protruding head (1) and a flush head (2) ready for installation.

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Step # 3:

PROTRUDING HEAD

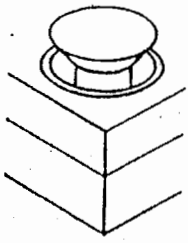
Check for adequate stack-up fillet relief.



NOTE:
Cutter for fillet relief of operation need not travel any deeper than necessary to permit full bearing of the flange with the structures as shown.

FLUSH HEAD

Check for countersink depth.

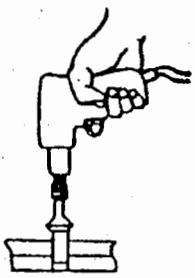


NOTE:
Sleevebolt is used as gage for proper countersink depth during drilling operation.

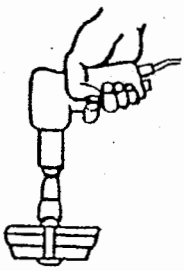
Step # 4:

No special tooling is required for installation, a rivet gun may be used or the bolt may be tapped in until the specified nut can be placed on the threaded end. Pull the bolt in place by tightening the mated nut.

STEP 4



Bolt driven in using standard rivet gun.



Bolt may be seated, after driving in part way, by applying torque to nut. Final desired torque should be applied to nut to complete the installation.

For final torque see Table III.

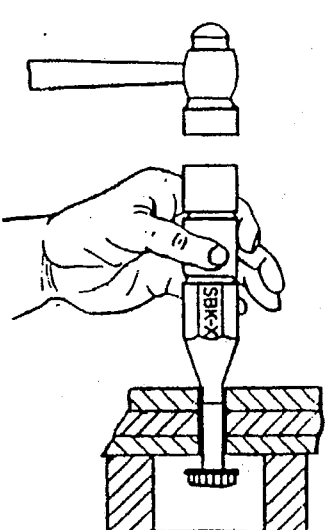
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BOLT REMOVAL:

Removal of the fastener is a two step process as follows:

Step 1.

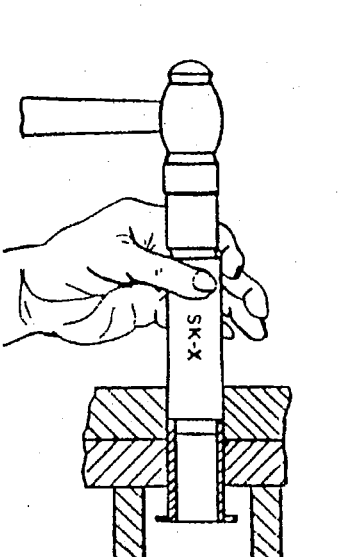
After removing the counterbore nut the pin is knocked out using the appropriate knock-out tool SBK-(x)*.
*x designates bolt thread size (available through PB Fasteners).



BOLT REMOVAL

Step 2.

After removing the pin the sleeve may be knocked out using the appropriate knock-out tool SK-(x)*.
*x designates bolt diameter (also available through PB Fasteners).



SLEEVE REMOVAL

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Licensee=Defense Supply Center/922(002100, User=McCarrick, John

TABLE I. DRILL CHART FOR FLUSH-HEAD BOLTS

Drill Size	Decimal Equivalent	Radius +.005 -.000
B	.2380	.040
6.20M	.2441	.040
1/4"	.2500	.040
F	.2570	.040
6.70M	.2638	.040
N	.3020	.040
5/16"	.3125	.040
O	.3160	.040
8.10M	.3189	.040
P	.3230	.040
Q	.3320	.040
U	.3680	.050
3/8"	.3750	.050
9.70M	.3819	.050
9.90M	.3898	.050
X	.3970	.050
11.00M	.4331	.050
11.20M	.4409	.050
11.40M	.4488	.050
11.50M	.4528	.050
11.60M	.4567	.050
11.80M	.4646	.050
1/2"	.5000	.060
13.00M	.5118	.060
13.20M	.5197	.060
17/32"	.5312	.060
13.80M	.5433	.060
14.50M	.5709	.060
14.75M	.5807	.060
15.00M	.5906	.060
15.25M	.6004	.060
15.50M	.6102	.060

NOTE 1: Drills referred to are combination drills with 100° countersink and have the appropriate listed radius.

NOTE 2: Sleevebolt Assembly is used as a gage for proper countersink depth during drilling operation.

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TABLE II. DRILL CHART FOR PROTRUDING HEAD BOLTS

Drill Size	Decimal Equivalent	45° Chamfer Diameter +.010 -.000	Radius +.005 -.000
B	.2380	.286	.035
6.20M	.2441	.292	.035
1/4"	.2500	.298	.035
F	.2570	.305	.035
6.70M	.2638	.312	.035
N	.3020	.350	.035
5/16"	.3125	.361	.035
O	.3160	.364	.035
8.10M	.3189	.367	.035
P	.3230	.371	.035
Q	.3320	.380	.035
U	.3680	.426	.040
3/8"	.3750	.433	.040
9.70M	.3819	.440	.040
9.90M	.3898	.448	.040
X	.3970	.455	.040
11.00M	.4331	.491	.040
11.20M	.4409	.499	.040
11.40M	.4488	.507	.040
11.50M	.4528	.511	.040
11.60M	.4567	.515	.040
11.80M	.4646	.523	.040
1/2"	.5000	.578	.050
13.00M	.5118	.590	.050
13.20M	.5197	.598	.050
17/32"	.5312	.609	.050
13.80M	.5433	.621	.050
14.50M	.5709	.649	.050
14.75M	.5807	.659	.050
15.00M	.5906	.669	.050
15.25M	.6004	.679	.050
15.50M	.6102	.688	.050
41/64"	.6406	.785	.083
21/32"	.6562	.800	.083
17.00M	.6693	.813	.083
43/64"	.6719	.816	.083

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TABLE II. DRILL CHART FOR PROTRUDING HEAD BOLTS (Continued)

Drill Size	Decimal Equivalent	45° Chamfer Diameter +.010 -.000	Radius +.005 -.000
11/16"	.6875	.832	.083
45/64"	.7031	.847	.083
18.00M	.7087	.853	.083
23/32"	.7188	.863	.083
18.50M	.7283	.872	.083
47/64"	.7344	.878	.083
19.00M	.7480	.892	.083
49/64"	.7656	.910	.083
25/32"	.7812	.925	.083
51/64"	.7969	.941	.083
13/16"	.8125	.957	.083
27/32"	.8438	.988	.083
55/64"	.8594	1.003	.083
7/8"	.8750	1.019	.083
22.50M	.8858	1.030	.083
23.00M	.9055	1.050	.083
59/64"	.9215	1.066	.083
15/16"	.9375	1.082	.083
61/64"	.9531	1.097	.083
63/64"	.9844	1.128	.083
1"	1.0000	1.144	.083
1 1/64"	1.0156	1.160	.083
1 3/64"	1.0469	1.191	.083
1 1/16"	1.0625	1.207	.083
1 5/64"	1.0781	1.222	.083
1 1/8"	1.1250	1.269	.083
1 9/64"	1.1406	1.285	.083
1 5/32"	1.1562	1.300	.083
1 3/16"	1.1875	1.332	.083
1 13/64"	1.2031	1.347	.083
1 7/32"	1.2188	1.363	.083
1 9/32"	1.2812	1.425	.083
1 19/64"	1.2969	1.441	.083
1 5/16"	1.3125	1.451	.083
1 11/32"	1.3438	1.488	.083
1 23/64"	1.3594	1.503	.083
1 3/8"	1.3750	1.519	.083
1 13/32"	1.4062	1.578	.097
1 27/64"	1.4219	1.594	.097

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TABLE II. DRILL CHART FOR PROTRUDING HEAD BOLTS (Continued)

Drill Size	Decimal Equivalent	45° Chamfer Diameter +.010 -.000	Radius +.005 -.000
1 7/16"	1.4375	1.610	.097
1 31/64"	1.4844	1.656	.097
1 1/2"	1.5000	1.672	.097
1 33/64"	1.5156	1.688	.097
1 35/64"	1.5469	1.719	.097
1 9/16"	1.5625	1.735	.097
1 37/64"	1.5781	1.750	.097
1 5/8"	1.6250	1.797	.097
1 41/64"	1.6406	1.813	.097
1 21/32"	1.6562	1.828	.097
1 11/16"	1.6875	1.860	.097
1 45/64"	1.7031	1.875	.097
1 23/32"	1.7188	1.891	.097
1 49/64"	1.7656	1.938	.097
1 25/32"	1.7812	1.953	.097
1 51/64"	1.7969	1.969	.097

CAUTION: NOTE: to clear the bolt radius and obtain complete seating of the head, drill holes with either a 45° chamfer to the size indicated, or drill holes with a radius to the size indicated.

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TABLE III.
RECOMMENDED MAXIMUM TORQUE VALUES

Dash No.	Flush Shear Bolt Min Shear Stress Level			Tension Bolt Min Tensile Stress Level		
	95 KSI	108 KSI	132 156 KSI	160 KSI	180 KSI	220 260 KSI
	Recommend Maximum Torque Values (In-Lbs)					
3	32	40	45	48	60	70
4	68	85	95	92	115	140
5	112	140	175	148	185	230
6	160	200	230	268	335	410
7	240	300	340	432	540	670
8	364	455	510	644	805	1000
9				864	1080	1340
10				1248	1560	1940
11				2344	2930	3650
14				3472	4340	5400
16				4880	6100	7600

Installation torque values for each dash number may be obtained by:

1. Read and record seating torque

2. Installation torque equals seating torque plus 10% to 15% of the seating torque.

Custodians:
Navy - AS
Air Force - 11
Army - AV

Preparing Activity:
Navy - AS
Project No. 53GP-0149-08

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